

NAVIN KARTHIK R

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Summary

Final-year Electronics and Communication Engineering student with strong problem-solving and analytical thinking, backed by programming fundamentals and data structures & algorithms practice in Java. Proven communication and presentation skills demonstrated through hackathon pitches and technical symposium presentations, with project experience spanning full-stack web development and embedded systems. Seeking an entry-level engineering role to build a career in technology through hands-on learning and consistent growth.

Education Details

Velammal Engineering College, Chennai B.E., Electronics and Communication Engineering – CGPA: 8.00	2023 – Present
Higher Secondary Certificate (HSC), Alpha Matriculation Higher Secondary School Percentage: 77.17%	Completed 2022
Secondary School Leaving Certificate (SSLC), Alpha Matriculation Higher Secondary School Percentage: 70%	Completed 2020

Internship Experience

Tamil Nadu Power Generation Corporation Limited – North Chennai Thermal Power Station, Stage-II <i>Internship Trainee (19 working days)</i>	Nov 20 – Dec 10, 2025
Studied thermal power generation processes, plant operations, and SCADA-based monitoring and control systems at a live thermal power station	

Project Details

SEED – Screening for Early Evaluation & Detection Team Project <i>AI-powered early screening platform; frontend and edge AI contributor, team ONE PIECE</i>	HACKHAZARDS '26
- Built frontend interfaces including the marketing landing page and authentication flows using React, plus a 3D data visualization module using Three.js - Integrated edge AI pipeline components (OpenCV, MediaPipe, XGBoost), aligned to ABDM compliance requirements - Collaborated in a 3-member team; achieved Global Rank 51-100 (Top 100) at HACKHAZARDS '26	
LOCO Hazard Detection Helmet Team Project <i>Microcontroller-based embedded safety system for real-time hazard detection and alerting</i>	
- Integrated GPS, DHT11 (temperature/humidity), and MQ2 gas sensors on an Arduino/ESP32 platform for real-time location and hazard tracking - Designed conditional alert logic for health monitoring (heart rate, blood oxygen/SpO2) and hazard sensors, triggering buzzer/emergency-switch responses and Telegram bot alerts over Wi-Fi - Debugged sensor data flow end-to-end, with live readings on LCD alongside remote alerting - Won 1st Prize at the IETE Symposium; presented at National Symposium 2K25, Velammal Engineering College	
RCCP – Reverse Chain Compliance Platform Team Project <i>Pharma drug-disposal reverse logistics and re-entry fraud detection platform</i>	24-Hour Hackathon, Saveetha Engineering College
- Designed a four-layer platform (retailer, distributor, manufacturer, regulator/public) on a shared append-only batch ledger keyed by manufacturer ID and batch number - Built deterministic fraud-detection logic using invariant checks (mass balance, batch existence, return conservation, certificate ceiling, expiry validation) instead of an ML-based approach, including a public no-login verification endpoint - Competed in a 24-hour Hackathon at Saveetha Engineering College against approximately 100 participating teams	

Strong Subject

Digital Electronics (DE)

Technical Skills

- Programming Languages: Java, JavaScript
- Web / Database: React, HTML, CSS, SQL (Basics)
- Embedded Systems: Arduino, ESP32, sensor interfacing
- Tools & Core Areas: Git, GitHub, Data Structures and Algorithms, Problem Solving, Debugging
- Soft Skills: Communication, Presentation, Teamwork, Adaptability

Achievements & Certifications

- Programming in Java – NPTEL Certification- 85%
- Global Rank 51-100, HACKHAZARDS '26 Hackathon -- Team ONE PIECE
- 1st Prize – LOCO Hazard Detection Helmet, IETE Symposium
- Hackathon Elearnix 4.0, Velammal Engineering College – Cleared Round 2 (Top 50 of 250 teams)
- INSTERA 2K25 – Project Presentation
- National Level Project Competition 2K25, Sri Sairam Engineering College – Participation
- IETE Student Forum Certificate – Institution of Electronics and Telecommunication Engineers